

Exotics

(3. Incorporating Dividends)

Introducing Dividends

When stock pays dividends, it changes the pricing equations, call and put bounds, and early exercise strategies for American options as well. Dividends can be specified in the following three ways;

- ▶ Continuous dividend yield (λ) → ^{q}
- ▶ discrete dividend as % of stock price (δ) - δ
- ▶ discrete dividend \$ amount (D) →

Dividends cause stock prices to drop by the same amount. While the stock-holder gets the dividend which compensates for the price drop, the option holder does not get the dividend, and it impacts the value of the option.



The dividend models mind map

Model 1 : ✗
continuous dividend
yield λ

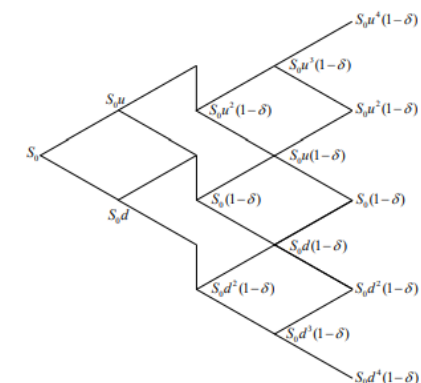
Black Scholes

$$S_0 \leftarrow S_0 e^{-\lambda T} \quad \checkmark$$

Binomial Tree:

$$S_0 \leftarrow S_0 e^{-\lambda T} \quad \checkmark \quad \text{(European)} \quad \text{(American)} \quad \text{✗}$$

$$q = \frac{e^{(r-\lambda)\Delta t} - d}{u - d} \quad \checkmark \quad \text{(European)} \quad \text{(American)} \quad \checkmark$$

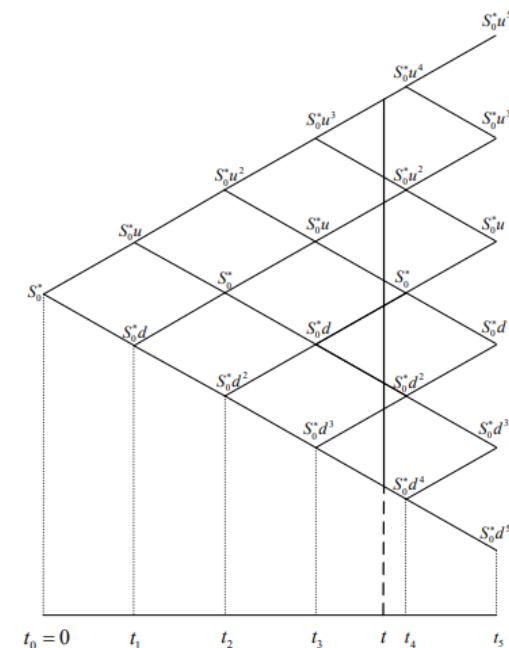


Model 2 :
discrete dividend (δ) as
% of stock price at τ

✗

1

✓ ✓
(European) (American)



Model 3 :
discrete \$ dividend
(D) at t

$$S_0 \leftarrow S_0 - D e^{-rt} \quad \checkmark$$

$$S_0 \leftarrow S_0 - D e^{-rt}$$

✓ ✗
(European) (American)

✓ ✓
(European) (American)

2

Model 1 : Continuous dividend yield λ (Closed Form)

The SDE can be updated to

$\uparrow r \downarrow \lambda$

$$\frac{dS}{S} = (r - \lambda)dt + \sigma dW \quad \rightarrow \quad S_T = S_0 \exp \left[\left(r - \lambda - \frac{1}{2} \sigma^2 \right) T + \sigma W_T \right]$$

$\mathbb{E}(e^{-rT} \times \text{payoff})$

$$= S_0 e^{-\lambda T} \exp \left[\left(r - \frac{1}{2} \sigma^2 \right) T + \sigma W_T \right]$$

$S_0 \rightarrow S_0 e^{-\lambda T}$

Black Scholes Adjustments

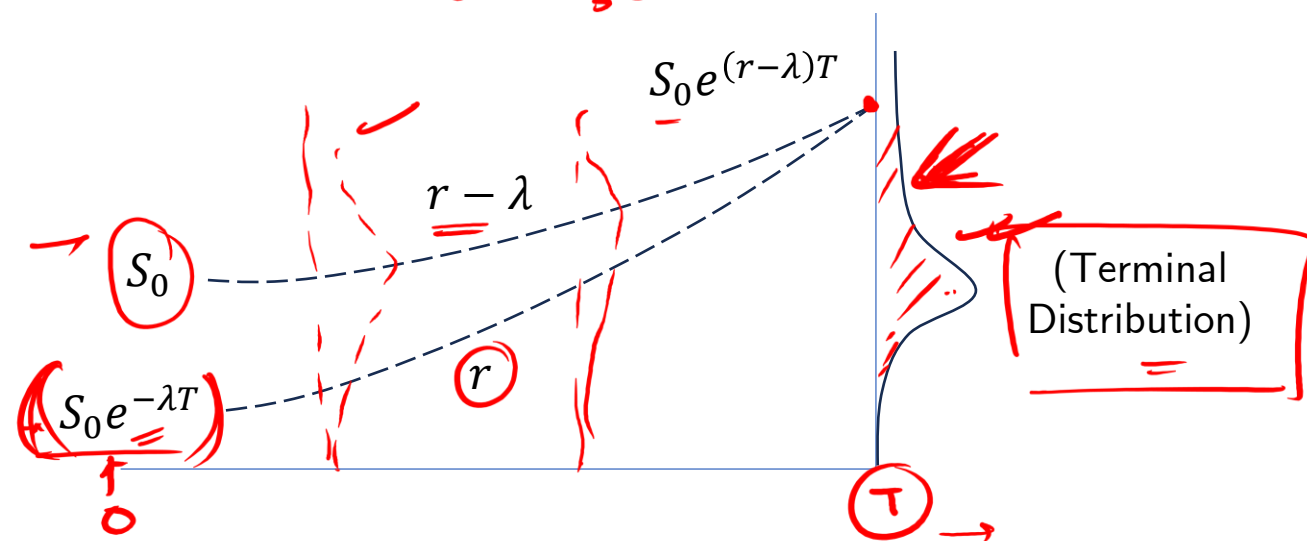
$$S_0 \leftarrow S_0 e^{-\lambda T}$$

$$\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + (r - \lambda) S \frac{\partial V}{\partial S} - rV = 0$$

$\downarrow$ drift $\downarrow$ discounting

$$d_{\pm} = \frac{\ln \left(\frac{S_0 e^{-\lambda T}}{K} \right) + \left(r \pm \frac{1}{2} \sigma^2 \right) T}{\sigma \sqrt{T}}$$

$$C_0 = S_0 e^{-\lambda T} N(d_+) - K e^{-rT} N(d_-)$$



Model 1 : Continuous dividend yield λ (Binomial Tree)

For Binomial Tree based implementation, we have two choices

- 1) $\otimes$ Substitute $S_0 \leftarrow S_0 e^{-\lambda T}$ (will work for European options, but not American Option. Why ?)
- 2) Update the risk-neutral probabilities to ensure stock price grows at $(r - \lambda)$. However, Option price using backward induction needs to be discounted using risk-free rate r . The adjusted risk-neutral probabilities are given as

$$\begin{aligned}
 \cancel{u}S\tilde{q} + d\cancel{S}(1 - \tilde{q}) &= \cancel{S}e^{(r-\lambda)\Delta t} \quad \Rightarrow \quad \tilde{q} = \frac{e^{(r-\lambda)\Delta t} - d}{u - d} \\
 \tilde{q}(u - d) &= e^{(r-\lambda)\Delta t} - d
 \end{aligned}$$



Early Exercise of American Call ?

We know that it's never optimal to exercise American Call early if the stock does not pay any dividend. For proof, refer to put call parity for European Options

$$\underbrace{P(t)} + S(t) = \underbrace{C(t)} + Ke^{-r(T-t)} \Rightarrow \boxed{C(t)} \geq \underbrace{S(t) - Ke^{-r(T-t)}}_{\substack{\text{SA} \\ \text{C}(+)}}$$

$\oplus$ SA
 $\downarrow K$ C(+)
 $\uparrow S$

But early exercise would mean $C(t) = \boxed{S(t) - K}$, so holding the call option is worth more than exercising it. In other words, the holder would rather sell the option than exercise it and get paid less.

With Stocks paying dividends, the put call parity for European Options becomes,

$$\underbrace{P(t)} + \underbrace{S(t)e^{-\lambda(T-t)}}_{\text{SA}} = \underbrace{C(t)} + \underbrace{Ke^{-r(T-t)}}_{\text{SA}} \Rightarrow \boxed{C(t)} \geq \underbrace{S(t)e^{-\lambda(T-t)} - Ke^{-r(T-t)}}_{\text{SA}}$$

So now we are comparing $\boxed{S(t) - K}$ against $\boxed{S(t)e^{-\lambda(T-t)} - Ke^{-r(T-t)}}$. Early exercise is a possibility if λ is larger.

$\lambda \rightarrow 0$
 $\lambda \uparrow \pi$



Early Exercise of American Put ?

We know that it might be optimal for American put to exercise early even if stock does not pay dividends. In the presence of dividends, the possibility of an American put to exercise early diminishes.

$$\begin{aligned} & \downarrow \\ & \text{call}(K - \underline{S}) \\ & \downarrow \\ & \text{call}(Ke^{-rT} - S) \end{aligned}$$



American Call Inequality

For a stock paying dividends, we have the following inequality

$$\begin{aligned}
 \checkmark \quad & \underline{C_{bs}}(S_t, t, K, T) \leq \underline{C_{am}}(S_t, t, K, T) \leq \underline{C_{bs}}(S_t, t, K, T) + \underbrace{S_t(1 - e^{-\lambda(T-t)})}_{\text{(American Options benchmarked against European counterparts)}} \\
 \checkmark \quad & \underline{P_{bs}}(S_t, t, K, T) \leq \underline{P_{am}}(S_t, t, K, T) \leq \underline{P_{bs}}(S_t, t, K, T) + \underbrace{K(1 - e^{-r(T-t)})}_{\text{(American Options benchmarked against European counterparts)}} \\
 \quad & \underline{S_t e^{-\lambda(T-t)} - K} \leq \underline{C_{am}}(S_t, t, K, T) - \underline{P_{am}}(S_t, t, K, T) \leq \underline{S_t - K e^{-r(T-t)}} \quad \text{(Put-Call Inequality)}
 \end{aligned}$$

Handwritten notes: A red circle around S_t and K in the third equation. Red arrows pointing to the S_t and K terms in the first two equations.

Notice that if $\lambda = 0$, then

$$\underline{C_{bs}}(S_t, t, K, T) \leq \underline{C_{am}}(S_t, t, K, T) \leq \underline{C_{bs}}(S_t, t, K, T)$$

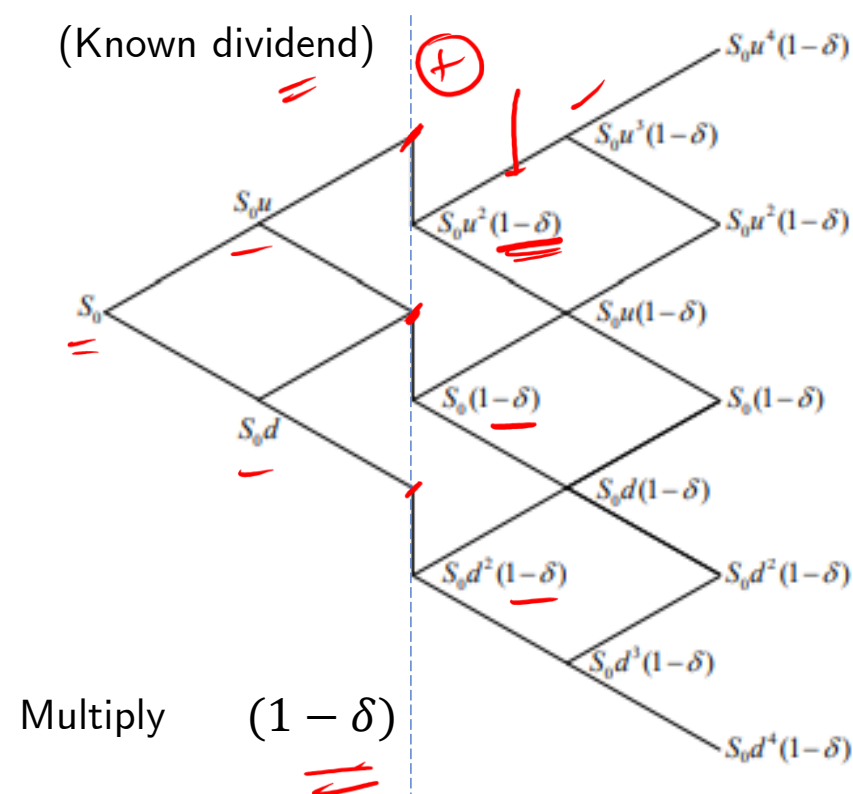
$$\Rightarrow \underline{C_{bs}}(S_t, t, K, T) = \underline{C_{am}}(S_t, t, K, T)$$

Model 2 : Known Discrete dividend δ as % of Stock

So, in this model, we have a known dividend δ on a future date expressed as % of stock price. There is no adjustments to make the Black Scholes closed form formula work. We can use the binomial tree.

The idea is to shift the entire tree downward by a factor of $(1 - \delta)$ on the day of dividend.

$\delta\%$



Model 3 : Known Discrete dividend dollar amount D

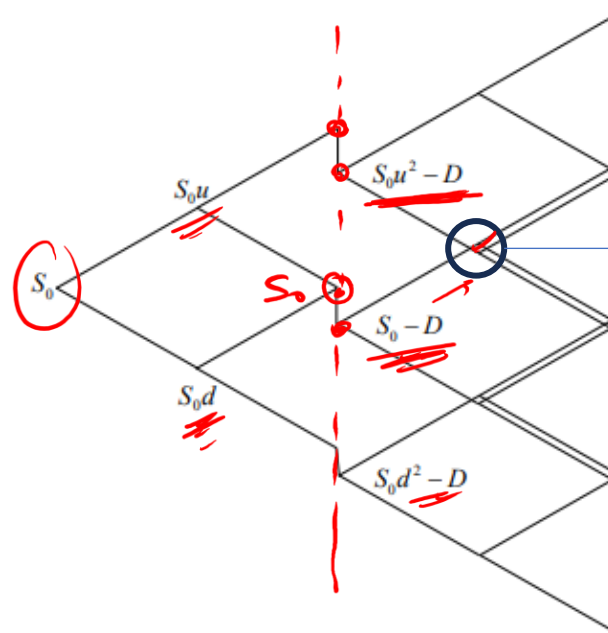
So, in this model, we have a known dividend dollar amount D on a future date $t < T$

❑ For Black Scholes, replace S_0 with $S_0 - De^{-rt}$

❑ For Binomial Tree Model

Method 1 : replace S_0 with $S_0 - De^{-rt}$ (works for European Option, but not for American Option)

Method 2 : Deduct dividend amount D at t . (Only problem is tree won't be re-combining after that)



$$(S_0u^2 - D)d \neq (S_0 - D)u$$

$$\begin{array}{cc} S_0u^2d - Dd & S_0u - Du \\ (S_0u^2 - D)d & (S_0u - D)u \end{array}$$

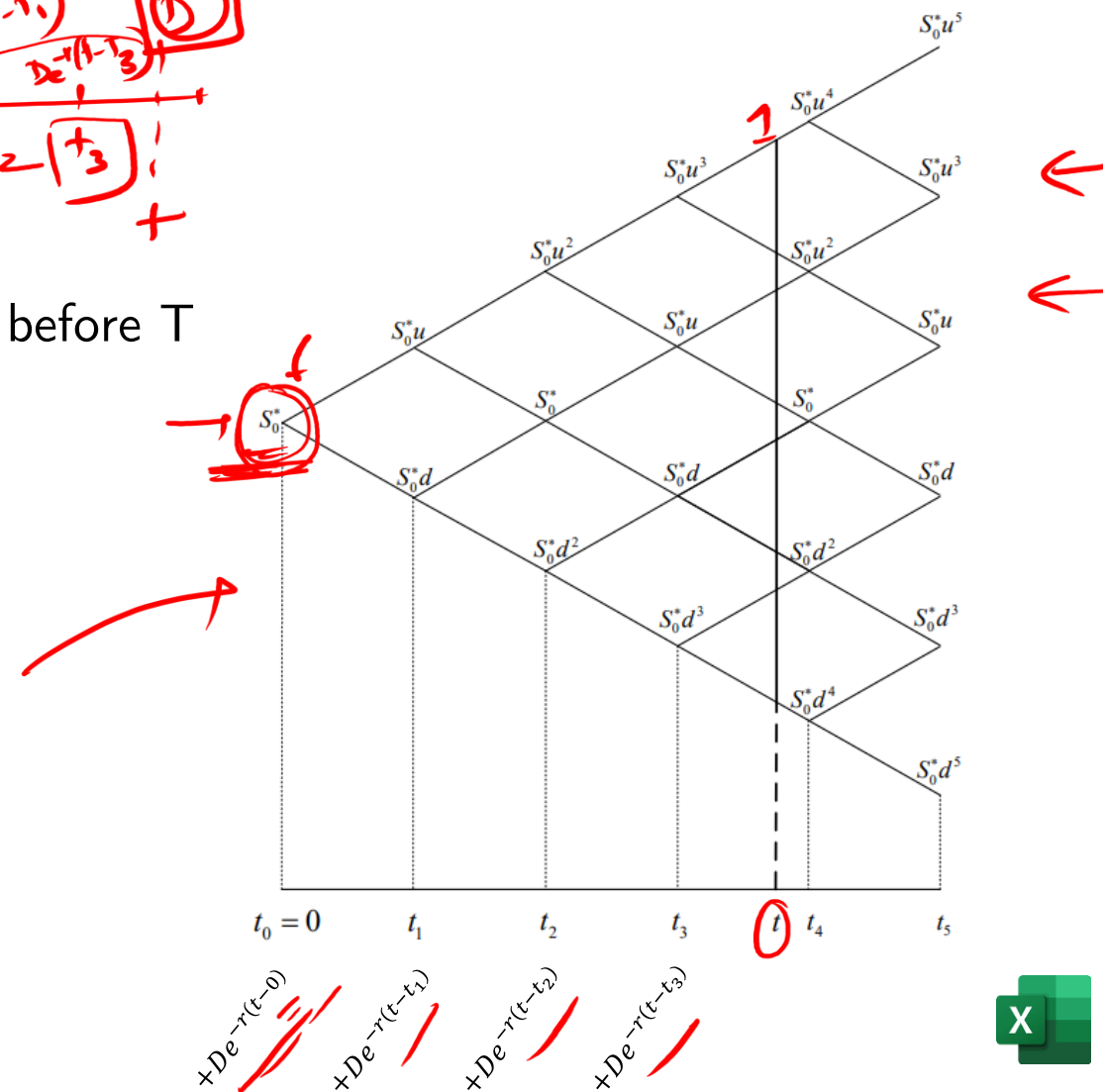
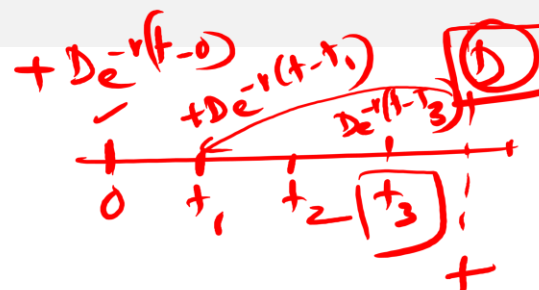
Model 3 : Known Discrete dividend dollar amount D

Method 3 :

Replace S_0 with $(S_0^*) = S_0 - \underline{\underline{De^{-rt}}}$ ✓

Grow the tree as usual (recombining)

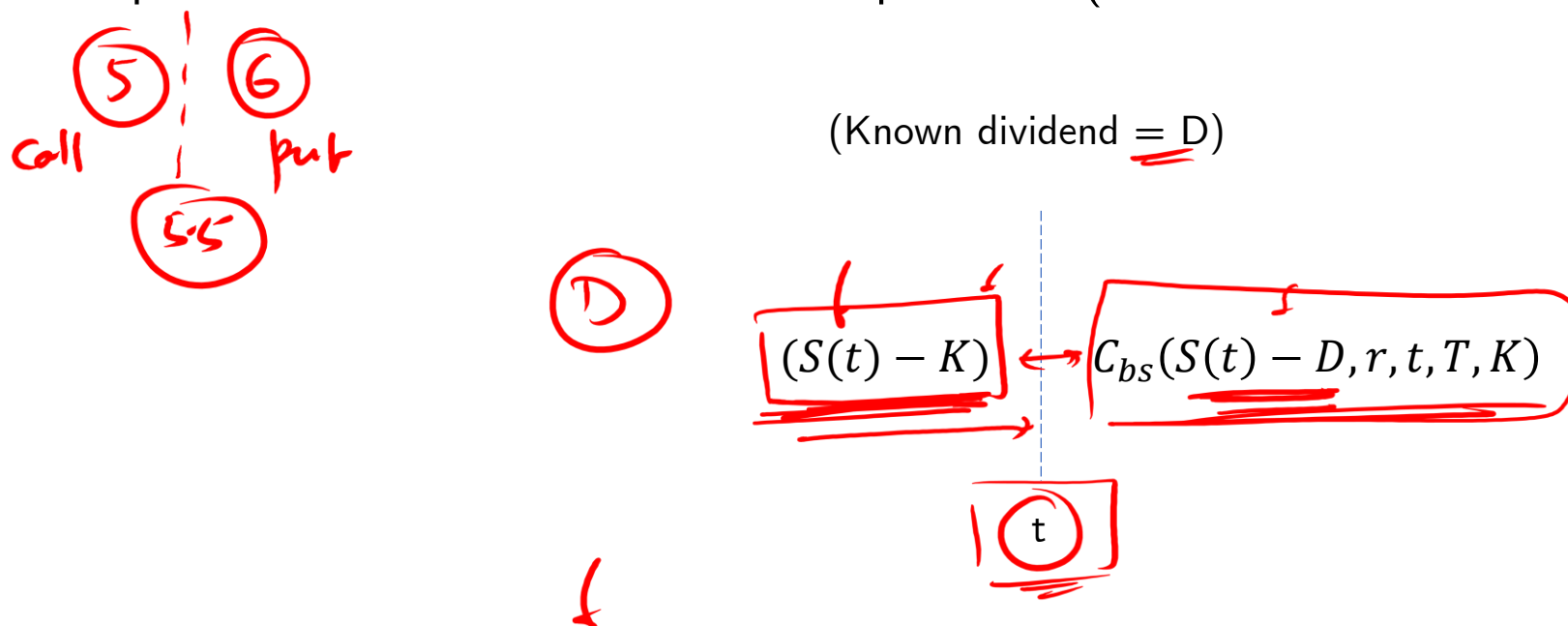
Add the PV of dividends back to the respective nodes before T



When early exercise makes sense ?

In the presence of dividends, the American put will ideally wait to exercise right after the dividend.

The American call option ideally should not be exercised early, but if the dividend is large enough, it makes sense to wait until just prior to the dividend paid. This is because right after, the stock price will drop and then it will behave like European call (no need to exercise until expiry)



Exercise if $(S(t) - K) > C_{bs}(S(t) - D, r, t, T, K)$

Assignment



1. Future Options ✓ (S)
- a. Derive the Black Scholes PDE for an Option on Future. Hint : $F(t, T) = S(t)e^{r(T-t)}$ ✓ $\frac{dS}{S} = rdt + \sigma dW$
 $\frac{dF}{F} = rdt$
- b. How can we tweak Black Scholes formula for vanilla options to make it work for Future Options ?
- c. Calibrate risk neutral probability in binomial tree setting to value Future Options.

2. Dividends on multiple dates ✓

In the example done in the class, instead of 1 dividend, assume there are 2 dividends announced (\$10 at 5.5 months and \$10 at 8.5 months). Answer the following questions

- a. How can we make Black Scholes work for European options with the 2 dividends.
- b. Build the binomial tree and calculate prices of European and American Call and Put maturing in 1 yr.